

Mice navigate scent trails using predictive policies

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Abstract

Animals actively sense their environment to extract features of interest to guide behaviors. For mammals, odors are prominent environmental features which are sampled by active modulation of sniffing and orofacial orientation. We sought to understand the strategies that mice use to navigate surface-bound odor cues. We presented mice with dynamic, non-repeating odor trails using a paper treadmill, and observed their behaviors as they collected rewards offered randomly along the trail. By combining high-speed videography over long distances with quantitative behavioral analyses, we find that mice rapidly learn to track odor trails persistently and precisely. Mice with a single nostril blocked can track odor trails, but with a lateral bias and lower precision than control animals. Tracking is severely impaired in animals with both nostrils intact but with interhemispheric communication disrupted by anterior commissure transection. Respiration measurements revealed that a sniff close to the trail triggers a rapid turn towards the trail, a reaction that is lost in commissure-cut animals. Importantly, trail tracking is not simply reactive but involves adaptation to and retention of a short-term memory of the trail geometry and statistics. Our results, recapitulated by a Bayesian inference model, indicate that mice combine immediate sensory information with an internal model of the odor environment to follow odor trails efficiently.

Introduction

Animals use their chemical senses to interact with and explore the world. Odor-based navigation is well known in insects, birds and mammals – notably rodents and dogs – but the behavioral strategies employed remain poorly understood. By making a series of decisions using stimuli in the world and using an internal map of the environment, animals can navigate large distances. Scent trails act as navigational landmarks that enable animals to identify conspecifics and salient places like home (Draft et al., 2018; Jones and Urban, 2018; Khan et al., 2012; Porter et al., 2007; Thesen et al., 1993). However, following such trails is a nontrivial problem because they are intermittent and dynamic. Furthermore, olfaction is a proximal sense. In contrast to vision-based navigation, where distant targets and trajectory options are immediately apparent, olfactory cues provide no obvious information about the spatial distribution or future locations of the chemical trail.

To track scent trails on the ground or odor plumes in air, animals make characteristic zig-zag like movements, commonly called casting (Draft et al., 2018; Khan et al., 2012; Porter et al., 2007). Such casting movements are observed in flying insects and are thought to be a part of search strategies employed to discover the source (Basu and Nagel, 2024; Cardé, 2021; Reddy et al., 2022a). Moths are known to complement casting with upwind surges upon detecting odors. Such search strategies are complemented by other sensory modalities, including optomotor anemotaxis (Álvarez-Salvado et al., 2018; Demir et al., 2020; Gomez-Marín and Louis, 2014).

Surface-borne scents pose a distinct set of challenges compared to airborne odors, where dynamic olfactory landscapes arise even when plumes are continuous, and more so when they are discontinuous due to turbulent transport (Gorur-Shandilya et al., 2019; Reddy et al., 2022a). Scents adsorbed on surfaces have a more stable distribution and have far slower transient temporal fluctuations than airborne cues. However, surface-borne scents can degrade over time, may not have predictable spatial gradients, and are challenging to track when they are meandering and discontinuous. Despite these challenges, wolves, dogs and bears can track trails many miles in length with remarkable accuracy (Peters and Mech, 1975; Thesen et al., 1993; Togunov et al., 2017). To follow a scent trail, an animal must conduct an initial search to locate the scent, followed by estimation of the heading of the trail, and direct subsequent movement to track the trail. In laboratory settings, animals as diverse as rodents, ants, and humans show similarities in tracking strategies despite differences in sensorimotor constraints (Draft et al., 2018; Jones and Urban, 2018; Khan et al., 2012; Porter et al., 2007).

Strategies for odor-guided animal navigation of airborne or surface-bound cues can be distinguished based on whether they are reactive or predictive (Emonet and Vergassola, 2024; Murthy, 2024; Reddy et al., 2022a). In one formulation, animals react to immediate sensory cues or landmarks to make decisions – for example, surging upwind when encountering odors (Álvarez-Salvado et al., 2018; Demir et al., 2020; Gomez-Marin and Louis, 2014) or turning based on spatial or temporal gradients (Gardiner and Atema, 2010; Gaudry et al., 2012; Khan et al., 2012; Liu et al., 2020; Rabell et al., 2017). Many models of odor navigation rely on such reactive strategies (Belanger and Arbas, 1998; Hengeniuss et al., 2021; Li et al., 2001; Russell et al., 2003; van Breugel and Dickinson, 2014), which are effective when cues are frequent and reliably indicate direction. Indeed, previous studies have shown that rodents can track relatively simple odor trails in a laboratory setting, and have proposed reactive strategies involving comparisons of odor intensities across nostrils or successive samples, effectively calculating spatial or temporal gradients of odor (Khan et al., 2012; Liu et al., 2020). However, the strategies described previously rely on immediate sensorimotor responses, and do not work when odor stimuli are intermittent or absent. These reactive strategies also do not easily account for any adaptations or reliance of prior experience.

On the other end, navigation strategies based on predictive models of the environment make use of prior knowledge, including recent experiences (Jayakumar and Murthy, 2022; Reddy et al., 2022b; Singh et al., 2023; Vergassola et al., 2007). These predictive strategies are particularly helpful when sensory cues are sparse, conditions under which reactive strategies may fail during long pauses in cue encounters. In particular, a recent theoretical proposal offers a predictive framework where an agent can use past encounters with the trail to estimate the trail heading direction as well as an angular sector of uncertainty about the heading (Reddy et al., 2022b). This formulation is broadly related to Bayesian inference, in which agents can balance current sensory input and priors to estimate the posterior probability of the underlying cause (Aitchison and Lengyel, 2017). Whether and how mice or other animals use a working memory of the scent trail statistics to efficiently track uncertain trails, is unknown.

Here, we used a custom-designed paper treadmill apparatus to print long odor trails of varying geometry and combined it with quantitative behavioral analysis and modeling to show that mice combine immediate sensory information with memory as they navigate scent trails.

Results

Mice quickly learn to follow continuous ground-based trails

Using our custom-built paper treadmill (Figure 1a), we presented mice with continuous odor trails of 2-phenylethanol printed in different shapes and statistics (Sample videos in https://github.com/VNMurthyLab/trail_tracking/blob/main/README.md). The experimental setup uses near infrared lighting (>850nm) to enable video recording while rendering the environment largely invisible to mice. To encourage mice to follow odor trails for long periods of time, we presented chocolate flavored Ensure drops delivered randomly on the trail to increase their motivation (see Methods). Naïve mice followed odor trails even on the first day of exposure and progressively became experts at closely following trails with their snout by day 6, as evidenced by their reduced deviations (Figure 1b-e; 65.1 ± 3.67 % tracking in expert vs 7.0 ± 2.63 % in naïve, mean $\pm$ s.e.m, (9 animals) $p < 1e-16$, Kolmogorov-Smirnoff (KS) test). To select episodes of trail tracking, we used a threshold of staying within 2.5cm on either side of the trail for at least 2.5 seconds. Over days, such epochs of trail tracking increase in number and duration (Figure 1f-h; 3.3 ± 0.816 epochs in naïve vs 18.6 ± 1.269 in expert, mean $\pm$ s.e.m, 9 animals, $p = 0.0039$, Wilcoxon signed-rank test). The improved trail tracking metric with experience was not due to specific selection criteria (Figure S1a-c, 77.1 ± 2.19 % tracking in expert vs 10.1 ± 3.55 % in naïve, mean $\pm$ s.e.m, at 3.5cm for 2.5 seconds and Figure S2d-f, 39.3 % in expert vs 5.4 % in naïve at 1.5cm for 2.5 seconds, mean $\pm$ s.e.m).

We could influence how mice sample the odor by varying the speed of the paper in the arena. We found that trail tracking fidelity varies as a function of the trail width and paper speed (Figure S2a,b). However, the identity of the odor used had no effect on the fidelity of tracking (Figure S2c). Mice quickly learned to consume the rewards on the trail and by day 5 were able to collect ~70% of the stochastically presented rewards (Figure S2d,e). When tested with similar treadmill speed and reward statistics but without a trail, mice manage to collect < 30% of the overall rewards (Figure S2e), indicating that mice must follow the odor trail faithfully to obtain rewards at a high rate. In further support of this idea, mice collected significantly fewer rewards if the droplets were deposited away from the trail (Figure S2e).

By flexibly varying the odor printer's position over time, we could present trails of diverse shapes. Mice that were experts at following simple trails were much less accurate when presented with tortuous trails with frequent kinks (Figure 1i, j, S2f,g) (45.6 ± 4.90 % tracking in expert mice tracking straight trails as opposed to 16.8 ± 2.23 %, mean $\pm$ s.e.m, when presented tortuous trails, (5 animals) $p = 2.175e-11$, KS test). Mice lower their snouts to the ground to sample the trail, presumably to maximize the signal from volatile odors that

disperse quickly above the ground. We used triangulation from multiple cameras to characterize their snout movements in three-dimensional space (see Methods). The vertical distance from the snout to the paper was highly variable on the first day and became tightly distributed around the trail over a few days, indicating that mice start sampling the ground more than the air as they become better at following odor trails (Figure 1k-m). When the trail odor was also presented as an airborne cue, expert mice are poorer at following the trails, indicating that mice track the odor and not some texture or scent from the printer ink (Figure S3a, b).

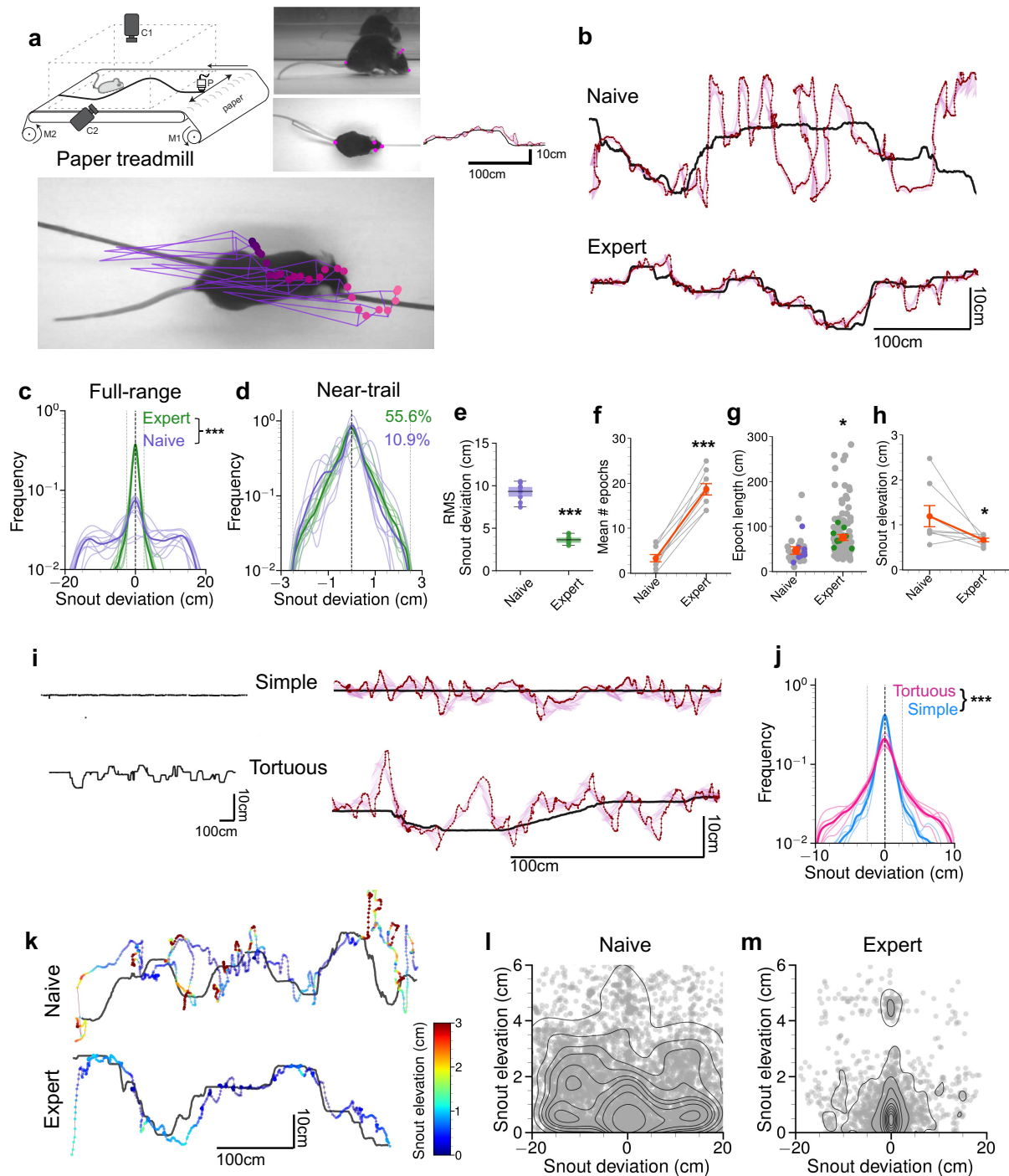


Figure 1: Mice rapidly learn to follow continuous trails. **(a)** Schematic of the custom-built paper-based treadmill. M1 & M2: rollers, with M1 coupled to a DC motor; C1 & C2: cameras; P: printhead that houses the printhead. The mice are restricted to the width of the chamber with walls made of acrylic that allow imaging with cameras. Odors are delivered using a printhead or a brush, controlled by stepper motors, allowing us to deliver varying geometries on the fly. Key points on the mouse (snout, right and left ear bases as well as base of the tail) identified using DeepLabCut (Mathis et al., 2018) are shown in magenta. **(b)** Examples showing the trajectory of a mouse on day 1 (naïve) and on day 6 (expert) following a trail. The trail is shown in black. The snout is shown in purple, with a rhombus joining the snout, ears and base of the tail in pink. **(c)** Distributions showing the snout deviation from 6 mice when naïve (day 1) vs when they have become experts (day 6), *** $p < 0.001$, Kolmogorov–Smirnov test. Each line indicates a unique animal. **(d)** The same data shown in panel c, but subsampled to include only points closer than 2.5cm on either side of the trail, numbers indicate percentage tracking. **(e-h)** Performance metrics of following a trail in naïve and expert mice. Each dot represents an individual animal, except in g where each colored dot represents an animal. **(e)** Root mean square deviations from the trail either in naïve or expert animals. *** $p < 0.001$, Two-tailed Wilcoxon rank-sum test. **(f)** Mean number of epochs, *** $p < 0.001$, Two-tailed Wilcoxon rank-sum test. **(g)** Length of each epoch, *** $p < 0.001$, Mann-Whitney U test and **(h)** Elevation of the snout from the paper floor, * $p < 0.05$, Two-tailed Wilcoxon rank-sum test. **(i)** Left: 2 example trails presented to mice. Right: Examples of a mouse following either a simple straight or a tortuous trail. **(j)** Distributions of snout deviations for 5 mice when following either a simple or a tortuous trail. **(k)** Examples of trail tracking in naïve or expert mice, with the vertical elevation of the snout mapped on to the tracks (colormap representing snout elevation shown to the right). **(l, m)** Distributions showing snout occupancy in 3d space in naïve mice **(l)** and when they have become experts **(m)**.

Overall, our experiments show that mice naturally follow odor trails but learn to improve their tracking fidelity over multiple days, and that trail tracking accuracy varies with the geometry of the trails encountered and the speed of the treadmill.

Bilateral inputs are helpful for accurately tracking odor trails

What are the strategies used by mice to track odor trails? Prior studies have suggested that bilateral signals arriving through the two nostrils aid in odor-guided navigation, including tracking an odor trail (Catania, 2013; Jones and Urban, 2018; Khan et al., 2012; Rabell et al., 2017; Rajan et al., 2006). These studies offered evidence for a role of bilateral olfaction using selective blockade of one nostril. After mice became experts at tracking odor trails in our set up, we stitched one nostril (5 mice for left and 4 mice for right) and monitored how they followed trails. Consistent with previous studies, we observed that mice with just one intact nostril were still able to follow trails, but with a lateral bias (Figure 2a,b, $p = < 1e-16$, KS test). The biases were on the same side of the stitched nostril (Figure 2c, and Figure S4a-c), such that mice placed their good nostril close to the trail and the closed nostril away from the trail. We also found that the number of tracking epochs and the length of each epoch decreased in nostril-

stitched animals, whereas the snout elevation was not significantly different from control (Figure S4d-f, 18.7 ± 1.640 epochs in sham as opposed to 13 ± 1.732 epochs, mean $\pm$ s.e.m. when the nostril was stitched, $p = 0.003906$, Wilcoxon signed-rank test).

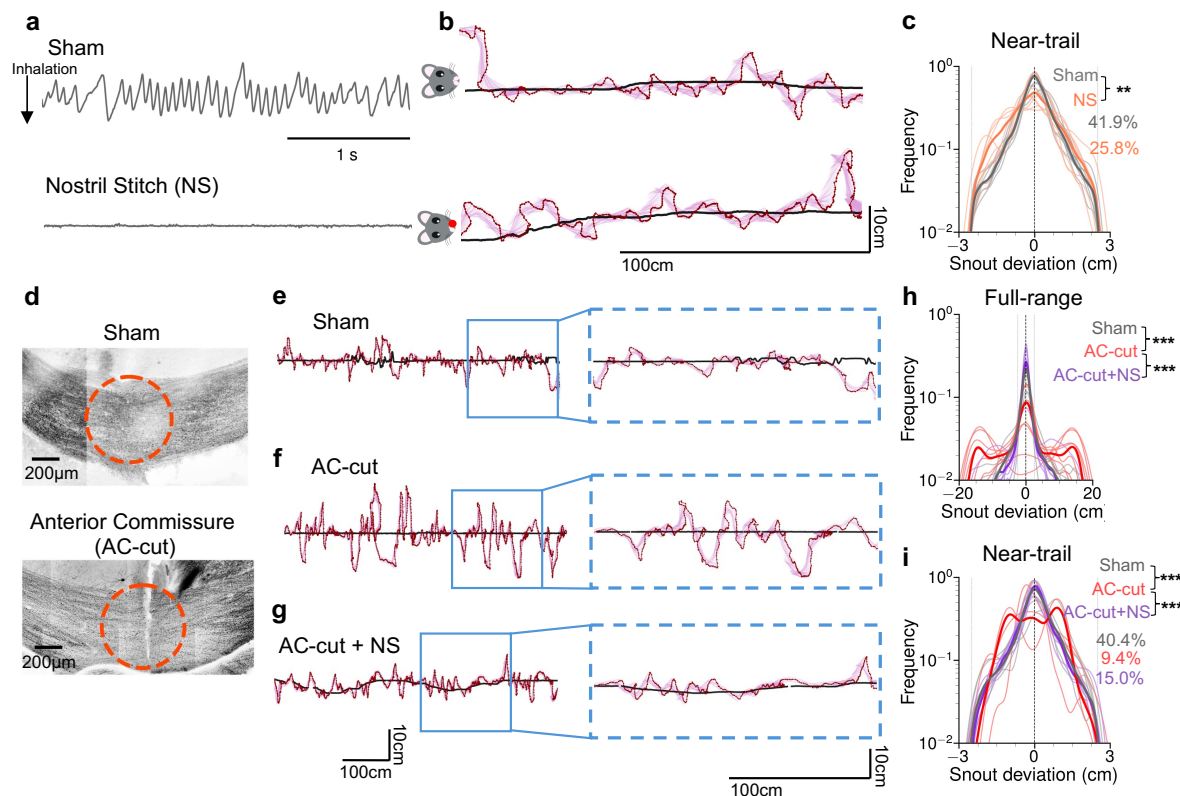


Figure 2: Coordination of bilateral inputs is helpful for accurately following trails. (a) Example time series from a thermocouple in a sham operated and a nostril stitched mouse. The thermocouple signal was at zero when the corresponding nostril was blocked. The arrow indicates inhalation. This was used to confirm that the nostril occlusion prevented airflow in that side. **(b)** Example trajectory from a mouse following a trail either after a sham stitch or when the left nostril was stitched. **(c)** Distributions of the deviations of the snout from the trail from either mice where there was a sham stitch or when the nostril was stitched, $**p < 0.01$, Kolmogorov–Smirnov test, numbers indicate percentage tracking. **(d)** Histological evidence for the transection of the anterior commissure. **(e-g)** Example trajectory from a mouse with either sham surgery, one with the anterior commissure transected and with an animal where the nostril was stitched post severing the anterior commissure. Inset shows zoomed in sections. **(h,i)** Distributions of snout deviations from at least 5 mice in conditions shown in (e,f,g) for all periods of behavior **(h)** or selected periods when mice were within 3cm of the trail **(i)**, $***p < 0.001$, Kolmogorov–Smirnov test. Numbers indicate percentage tracking.

Closing one nostril not only creates an asymmetry in stimulus acquisition but also likely reduces the total signal acquired. Additionally, this perturbation could lead to compensatory alteration of the strategies used for following trails, which might mask those used in unperturbed conditions. Therefore, we turned to a manipulation that leaves bilateral stimulus

acquisition intact but interferes with the communication between the two streams. We severed the anterior commissure, the main fiber bundle that connects the ventral olfactory regions in the two hemispheres, in particular the AON and piriform cortex (Brodal, 1948; Brunjes, 2012; Gurdjian, 1925; Kikuta et al., 2008; Schoenfeld and Macrides, 1984) (Figure 2d). Mice with severed anterior commissure follow trails poorly, taking more tortuous paths (Figure 2e, f). The number of tracking epochs and their duration as well as the snout elevation were significantly different upon cutting the commissure (Figure S5a-d; 15.7 ± 1.7 epochs in sham (5 animals), 2.6 ± 1.07 epochs, mean $\pm$ s.e.m., when the commissure was severed (7 animals), $p = 0.011$, Mann-Whitney U test). Interestingly, mice with their commissures transected failed to improve much over time (Figure S5e).

Careful inspection of the distributions of the snout deviations suggests that mice whose commissure was cut displayed a lateral bias toward one side of the trail or the other (Figure 2h). This suggested that there could be a form of rivalry between the two independent nostril inputs that remain unresolved in the absence of interhemispheric communication. To test whether there was such a rivalry, we stitched the nostril of the mice with their commissures cut. We observed that mice with their commissure cut, but with only one functional nostril, were able to follow an odor trail much more accurately, comparable to sham operated mice (Figure 2g-i, 23.4 ± 6.45 epochs when the anterior commissure was severed and one nostril blocked (5 animals) vs 15.7 ± 1.7 epochs in sham operated, mean $\pm$ s.e.m, $p > 0.6$, Mann-Whitney U test). This double manipulation also indicated that the commissure cut does not lead to some non-specific defects that cause animals to follow trails poorly. The behavioral rescue upon shutting one nostril is lost upon removal of the stitch.

These data indicate that, while mice can follow trails with a single intact nostril, bilateral information flow needs to be coordinated via the anterior commissure when both nostrils receive odor inputs.

Mice turn reactively after asymmetric trail encounters

Mice predominantly acquire information about the odor trail when they sniff, which can happen at different frequencies (Findley et al., 2021; Kepecs et al., 2006; Wesson et al., 2009). To determine whether and how mice modulate sniffing to aid efficient sampling and tracking, we measured respiration in mice following odor trails either using a thermocouple or using wireless diaphragmatic EMGs (Figure S6a-c). Wireless recordings had the important advantage of allowing free head movements, which may be difficult with tethers necessary for other methods.

Mice actively sampled the trail and displayed high rates of respiration when following the trail, in comparison to much lower rates when they were further away from the trail (Figure 3a,b). The respiration rate dropped to low levels when mice were consuming the rewards presented sporadically on the trail, in line with work on coordination of various orofacial rhythms that report low rate of breathing when swallowing (Kaku et al., 2025; Kleinfeld et al., 2023; Moore et al., 2013; Saito et al., 2003).

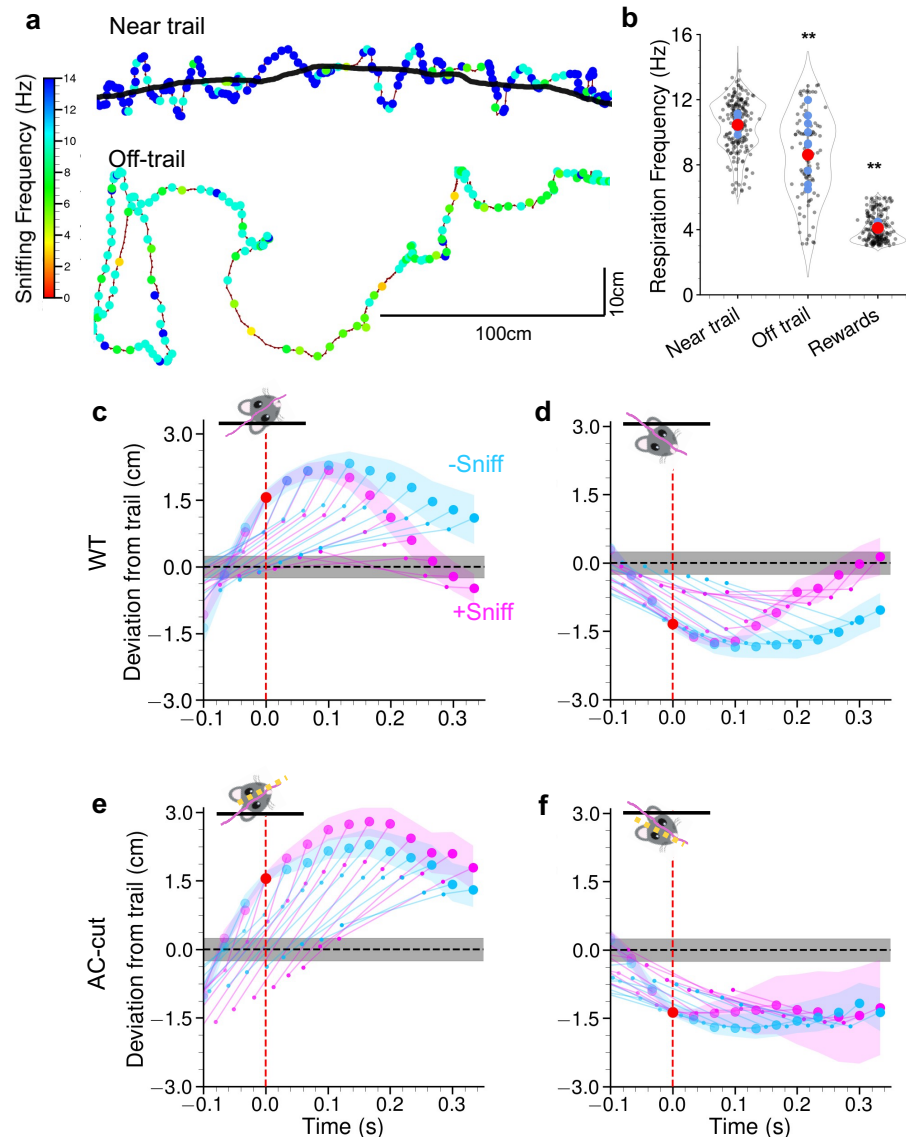


Figure 3: Modulation of respiration during trail tracking allows quick turns at trail edges. (a) Example trajectory of a mouse either following a trail or when there is no trail. Inhalations are shown in color. The colormap shows instantaneous respiration rate. (b) Average respiration rates in different conditions, **p<0.01, Kruskal-Wallis ANOVA with post hoc Mann-Whitney U tests using Bonferroni multiple comparisons correction. (c-f) Average mouse trajectories showing snout (large circles), shoulder and base of the tail, joined by lines. The red dot indicates the anchor point for the averages, when the mouse snout was within 1 - 2cm

of the trail after crossing it. Pink traces are averages when an inhalation was detected at $t = 0$, and cyan are when there is no inhalation at $t = 0$. Left and right panels are separate averages for trail crossings in opposite directions. Trail of 0.5cm is shown as a gray line. **(c, d)** is from wild type mice (at least 27 stretches from 12 mice) and **(e, f)** shows data from mice where the anterior commissure was transected (at least 28 stretches from 8 mice).

Previous studies (Catania, 2013; Jones and Urban, 2018; Khan et al., 2012), as well as our experiments with nostril occlusion and commissurectomy, suggest that mice may use bilateral comparison of odor intensity to track trails efficiently. We searched for signatures of such a computation in our data by asking whether asymmetric intensity signals across the two nostrils can trigger an orientation response. We computed the average trajectory of the snout and head of the mouse conditioned on whether or not there was a sniff when the snout was located at specific distances from the trail, from at least 25 stretches from at least 8 animals. An inhalation that occurs when the mouse is ~ 1.5 cm from the center of the trail as it is moving away from the trail triggers a reorientation of the head towards the trail within ~ 150 ms (Figure 3c). The head reorientation was followed by a full body turn. This rapid reorientation occurs much less prominently when there is no inhalation. Importantly, this reorientation does not occur after inhalation when the snout is much closer to the trail, such that the two nostrils are more symmetrically placed on the trail (presumably leading to similar odor intensities reaching the two nostrils) (Figure S6d,e).

If the reorientation is triggered by computations using the asymmetric sensory inputs to the two nostrils, disrupting the comparison of the two inputs should prevent this behavior. To test this idea, we repeated this analysis in mice with severed anterior commissure and found that there was no privileged reorientation triggered by inhalation (Figure 3e,f).

Collectively, these analyses suggest that upon inhalation, if there are differences in odor intensities in the two nostrils, mice orient towards the trail with the intensity comparison requiring interhemispheric communication through the commissure.

Mice use past trail encounters to estimate future heading

The reorientations triggered upon inhalation described above indicate that mice can use reactive strategies during trail tracking. It remains unknown whether and how mice use knowledge about trail features, internalized from prior experiences (long-term memory) or recent experiences (working memory or adaptation) to make decisions about their actions. Earlier experiments have indicated that rats cast with increasing amplitude upon losing the odor trail (Khan et al., 2012), which could be due to a simple expectation of the trail based on the preceding sniff. A recent theoretical model formulated a sector search strategy that is

based on a short-term memory of accumulated information from previous samples (which could result in presence or absence of odor) (Reddy et al., 2022b). We performed three separate experiments to test whether mice use recent experience to follow trails.

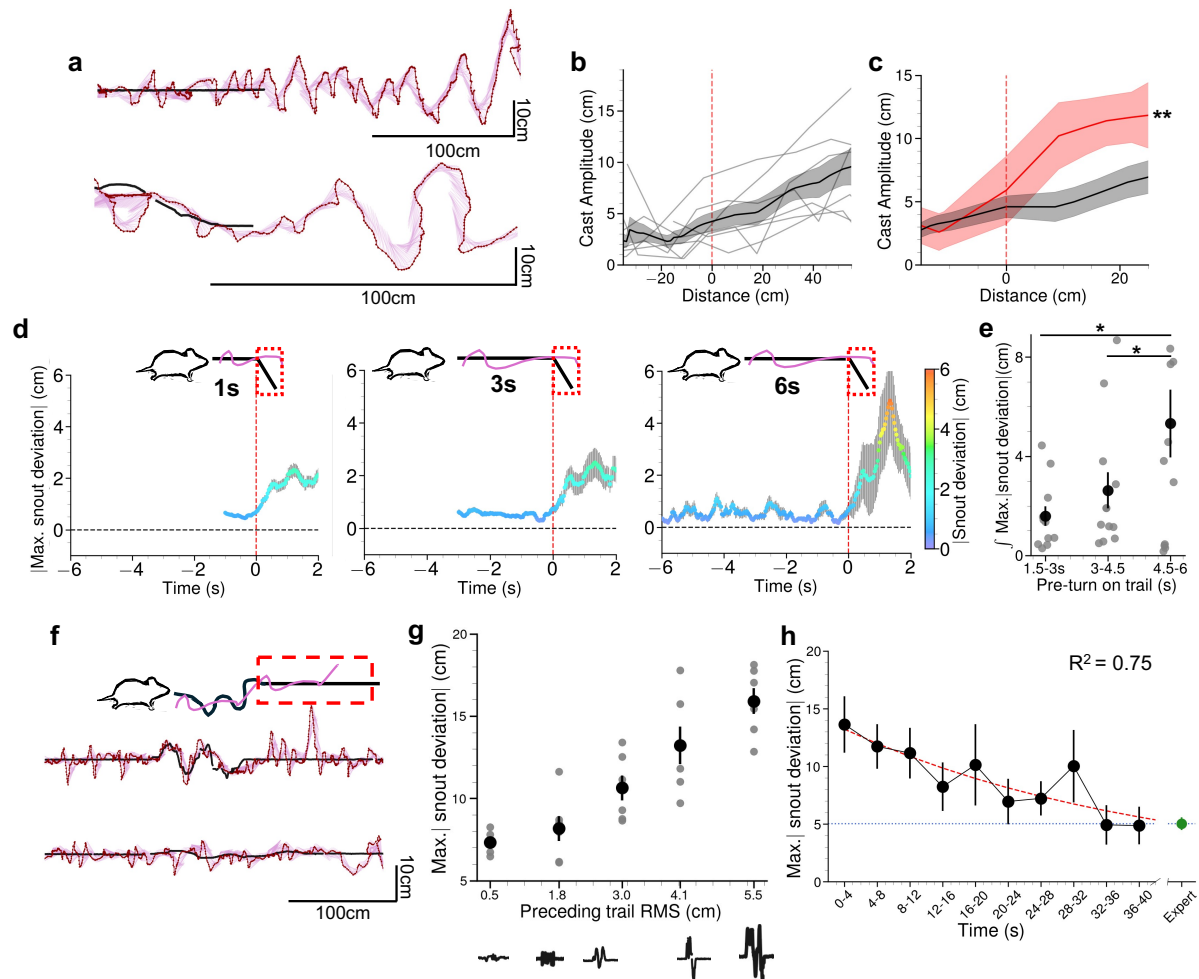


Figure 4: Mice use memory of previously encountered trail statistics. (a) Example trajectories from mice encountering a break while tracking a trail that is either straight (top) or tortuous (bottom). (b) Casting amplitude from 7 stretches from 5 mice encountering a break when following a straight trail. (c) Casting amplitude from 7 stretches from 5 mice encountering a break either when following a straight trail (black) or a more tortuous trail (red). (d) Snout deviations over time as mice encounter a sudden turn while following a trail. (e) Averages of the integrated maximum snout deviations over 2 seconds after encountering a turn after different periods of following a straight trail. (* p <0.05, Kruskal-Wallis ANOVA with post hoc Mann-Whitney U tests using Bonferroni multiple comparisons correction). Each point is for one session from one mouse. (f) Example trajectories from straight sections of a trail before or after encountering noise patterns in the trail (g) Maximum snout deviation in sections where the trail was straight preceded by trails of varying noise patterns. (h) Maximum snout deviation in a straight section of the trail immediately after encountering a trail with a root mean squared deviation of 4.1 cm. The average maximum snout deviation from 9 expert mice that have not faced noisy trails are shown in green. The R^2 value indicates goodness of fit, p -value=0.9032 (Chi-squared test). The best fitting time constant was 18.4 seconds.

First, we presented long straight trails, which were then interrupted by breaks. Upon encountering breaks mice perform casts of increasing amplitude (Figure 4a,b). Breaks introduced within complex tortuous trails lead to more rapidly growing casts by mice (Figure 4a,c), likely reflecting greater uncertainty in their trail heading estimates.

In a second set of experiments, we introduced unexpected sharp turns in the trail after varying periods of straight stretches. Mice make excursions away from the trail upon encountering sharp turns (Figure 4d, Figure S7a-c). If mice adapt to straight stretches, they might predict the future trail geometry to also be straight. If so, the length of experience in straight stretches would influence their behavior when encountering the unexpected turn. Indeed, when mice reach the turn after longer periods of tracking straight trails, they overshoot by a larger amount before correcting their heading (Figure 4e). This overshoot was not due to differences in forward velocity of the mice when reaching the turn (Figure S7d,e).

A third set of experiments involved the presentation of straight trails interspersed with periods of convoluted trails with different degrees of lateral excursions. In this design, mice will arrive at the straight stretches after periods of following trails with different convolutions. Remarkably, mice follow highly similar straight trails with very different accuracies, exhibiting much greater lateral deviations in the straight sections after following trails that are more convoluted (Figure 4f, g). To test how long the altered pattern of trail tracking persisted, mice were presented with one version of the convoluted stretch (with a root mean square deviation of 4.1cm) for 9-11s seconds, followed by a long straight segment. This experiment indicated that the altered trail tracking pattern returns to the baseline pattern of tracking a straight trail with a time constant of 18.4 sec (Figure 4h).

Taken together, these findings suggest that mice use an internal model of recently encountered trail statistics to estimate the trail heading direction and uncertainty, which informs the actions taken to follow the current trail.

Trail tracking models suggest extended memory is required for accurate predictions

Predictive search strategies use the history of past trail detections and internal models of the tracked trails to make inferences of their future heading. For instance, the animal may try to infer the future mean heading and its uncertainty to define an angular sector that it can move forward to search (Reddy et al., 2022b). Unveiling the details of the internal models used by mice is a major goal that extends well beyond the scope of our work here, but we can use the above data to compare simple reactive and predictive models. First, we reproduced the

model from Khan et. al. (Khan et al., 2012), which imposed automatic lateral movements of the nose, with turns that were modulated by spatial (across two nostrils) or temporal (across two sniffs) odor gradients. An internal estimate of the trail heading and error bounds were calculated with hand-crafted parameters that exploit up to the past five detections. Therefore, the model has a very limited memory with some heuristic “turning back” triggered by the nose going too far out of the heading estimate. An agent with these features and parameters can track fictive odor trails of simpler geometries similar to a purely reactive model (Figure S8a-d,i) without memory but fail to follow trails with gaps and is much worse on more complex trails with higher tortuosity (Figure S8e-h,j). The model also fails to recapitulate the history-dependent effects we observed in Figure 4d-g.

Therefore, for the agent to track efficiently under more difficult conditions, a reactive strategy alone is not sufficient, and a more extended temporal memory is needed. A Bayesian formulation of the problem of inferring the future trail heading (the posterior density) based on the current measurement (likelihood) and the previous accumulated knowledge of the trail from the past observations (prior) reads as follows

$$P(s_k | Y_{1:k}) = \frac{P(y_k | s_k, Y_{1:k-1}) P(s_k | Y_{1:k-1})}{P(y_k | Y_{1:k-1})}$$

Here, $Y_{1:k}$ is the set of all detections from the beginning of the hypothesized memory length up to the current time, *i.e.*, $Y_{1:k} = (Y_{1:k-1}, y_k)$ with y_k denoting the last measurement (sniff-related trail odor detection or non-detection). The variable s_k denotes the inferred state of the future trail heading, which could involve various statistics, such as mean heading, its uncertainty, curvature of the trail, frequency of breaks, etc. The first term in the numerator is the likelihood of a detection/non-detection, while the second is the inferred state of the trail prior to the current measurement. The denominator is a normalizing factor corresponding to the overall evidence. The goal of an agent tracking the trail is to estimate this posterior density (the left-hand term in the above equation) at each instance of decision making, which can be discretized at some acceptable resolution, and control its motion accordingly. A purely reactive strategy will rely entirely on the current measurement, *i.e.*, $P(s_k | Y_{1:k}) = P(s_k | y_k)$.

The use of prior knowledge is a key ingredient of classic state estimation procedures, such as Kalman filtering or the more general Bayesian particle filtering (Arulampalam et al., 2002; Chen, 2003; Särkkä and Svensson, 2023). Previous models of olfactory navigation have postulated an extended memory (Vergassola et al., 2007), or a more limited one (Khan et al., 2012; Reddy et al., 2022b), but it remains unclear whether and how mice (and other animals) use recent experience to obtain these estimates. In order to highlight the diversity of possible solutions to the problem and that it is quite open, we chose Gaussian Process Regression (GPR) as a general method to estimate the prior distribution with minimal assumptions (Arulampalam et al., 2002; Rasmussen and Williams, 2005) (see Methods). Our

Figure 5: An agent using memory reproduces many facets of tracking behavior. (a-d) Example trajectories of an agent using a Gaussian Process Regression to estimate the trail from previous contacts with the trail. The virtual trail is shown in black, the agent's trajectory in orange, the agent's trail encounters as sky blue dots, the agent's heading estimate in a sky blue line and the posterior uncertainty of the trail position ($\pm 2\sigma$, where σ is the standard deviation output by the GPR at each point) in gray cones. The example trajectories are from the agent following **a)** a simple straight line, **b)** a simple trail with breaks, **c)** curves of low tortuosity and **d)** curves of high tortuosity. **(e)** Deviations from the trail as shown by means $\pm$ s.e.m. are from 30 simulations, show that the agent is accurate in following trails. **(f and g)** Example trajectories of the agent encountering a break while following trails of two types of curvature. **(h)** Casting amplitudes measured from 30 simulations as shown by means $\pm$ s.e.m. show that the agent casts more widely when encountering a break following the trail of higher curvature. *** $p < 0.001$, Mann-Whitney U test. **(i and j)** Example trajectories from an agent following a straight section of a trail after encountering either low tortuosity **(i)** or high tortuosity **(j)**. **(k)** Snout deviations over time from agents following a trail. The deviations are from straight sections of the trail. The deviations are significantly higher when the trail was preceded by the curves of higher tortuosity. *** $p < 0.001$, Mann-Whitney U test.

We next tested the behavior of the agent upon encountering a break after having followed trails of either low or high tortuosity (Figure 5f-h). We find that the agent's heading is more uncertain when the break is preceded by higher tortuosity (Figure 5g compared to Figure 5f) and is reflected in the increasing casting amplitude (Figure 5h), resembling what we observed in mice (Figure 4a-c). Finally, we examined the effect of an agent with extended memory on trails that straightened after being convoluted as in Figure 4f. The agent qualitatively replicated the behavior of mice (Figure 5i-k). The trail tracking in the straight stretch has more lateral deviation when the agent has just been following a convoluted trail, since the uncertainty estimate continues to be influenced by the past experience within the memory window. The more convoluted the prior trail, the greater the lateral deviation in the straight stretch (Figure 5j, k).

Taken together, these results show that simple reactive or simple fixed rules fail to replicate key features of trail tracking observed in mice. In contrast, Bayesian models which use recent trail history to dynamically estimate heading direction and associated uncertainty reproduce observed mouse behaviors remarkably well.

Discussion

In this study, we used a continuously running paper treadmill to present mice with dynamic odor trails, allowing us to observe their behavior over extended distances and durations as they collected randomly placed rewards. Mice quickly learn to follow odor trails and become more accurate trackers over several days. While mice can follow trails with just one nostril,

bilateral integration is critical for accurately following trails when both nostrils are intact. Mice quicken their respiration when following trails to increase sampling rates, and asymmetric placement of their nostrils when a sniff occurs triggers a head turn towards the trail – a response absent in commissure-transected animals. Importantly, tracking an odor trail involved more than immediate reactions: mice retained short-term memories of trail geometry, as seen in their larger overshoots at sharp turns following longer periods of straight trails, and their adaptation to different trail patterns. A Bayesian model for inferring the trail properties using prior trail encounters with a memory lasting ~15 seconds could explain the mouse behavior well, whereas a purely reactive model or one with a simple fixed memory failed to explain the data. These findings suggest that mice integrate real-time sensory input with internal representations to efficiently track odor trails.

Our study was inspired by previous work in rats (Khan et al., 2012), and offers significant new insights into trail tracking behavior. First, by using indefinite, non-repeating trails, we avoided any memorization of the trail geometry by the animals. Instead, mice had to rely on the ongoing experience of the trail to guide their behavior. A second new finding was that mice can spontaneously track trails on the first day, and their improvement over days is mainly due to longer periods of engagement in tracking a trail as well as their 3D snout placement near the ground. Third, we show that nostril occlusion experiments, a mainstay of testing bilateral olfaction (Catania, 2013; Jones and Urban, 2018; Khan et al., 2012; Rajan et al., 2006), may have confounding effects that can be resolved by more nuanced manipulation of bilateral integration. Finally, by printing trails of many different geometries, we revealed previously unknown ability of mice to make inferences about the statistics of the trail. The ability to print long trails whose thickness, shape and integrity can be controlled dynamically, combined with modern markerless tracking and automated behavioral analysis (Mathis et al., 2018), facilitated the collection of extensive data under many different conditions. Such high throughput acquisition in a controlled laboratory environment allows a balance between stimulus control and presentation of highly variable, non-repeating segments that are hallmarks of naturalistic behavior.

One strategy proposed for detecting and remaining on the trail is bilateral comparison of odor intensities across the nostrils, which might help detect trail edges. Evidence for this idea has largely come from the effects of blocking one nostril. We confirm previous findings (Catania, 2013; Jones and Urban, 2018; Khan et al., 2012) that mice can still follow trails, but less accurately and with a lateral bias. This perturbation reduces significantly the overall information acquired by the animal and it is likely that animals adapt their navigational strategies to their new state. Therefore, we created a situation where mice acquire information

from both nostrils, but the interhemispheric communication between the two channels of information was disrupted by severing the anterior commissure. Mice were very poor at tracking in this condition, likely because information from the two channels could potentially be different (and conflicting), and without comparison may result in erroneous decisions which would take the animal away from the trail. Furthermore, the behavior does not improve over time. We note that the behavioral deficit is not due to some irreversible loss of navigational or other non-specific issues, since blocking one nostril in commissure-cut animals partially reverses the behavioral deficit.

The bilateral comparison idea was further supported by our analysis that revealed sniff-triggered head reorientation. Sniffs to one side of the trail within a short distance of ~1 cm trigger head reorientation towards the trail within 150 ms. The same sniff taken when the nostrils are much closer to the trail do not trigger reactive head turns, presumably because the odor intensity is very similar in both nostrils. Supporting such bilateral comparisons, we found that sniff-triggered reorientations were absent in commissure cut animals. Taken together, our data suggests that a sniff that leads to asymmetric activation of the two nostrils is detected in the brain through bilateral comparison and leads to head reorientation. There is clear evidence for reorientation of the nares towards an odor source presented asymmetrically to head-restrained animals, which is thought to be mediated by collicular circuitry (Rabell et al., 2017) – we predict similar neural circuits are involved in head reorientation in our behavior.

We found strong evidence that mice form a predictive model of the trail structure and do not track odor trails using a purely reactive strategy. First, how mice search for the trail upon encountering a break is influenced by the recent trail structure. If breaks occur after a period of linear (straight line) trail, mice search with a slowly expanding casting cone, as observed before in rats (Khan et al., 2012), and rationalized by the sector search theory (Reddy et al., 2022b). In contrast, if a break is encountered after a period of highly sinuous trail with unpredictable bends, mice search in a rapidly expanding casting cone. This dependence of the search sector angle on the previous trail statistics cannot be reconciled with a purely reactive strategy, which will only depend on current sampling. Instead, it requires predictive processing, and results from a greater uncertainty in the estimate of the trail heading direction (based on previous detections) when following a tortuous trail (Jayakumar and Murthy, 2022; Reddy et al., 2022b). A second set of observations comes from introducing unexpected turns in the trail after mice were following a long linear trail. If mice were on the linear section for a brief period, they deviated only a little when reaching the bend in the trail before correcting themselves. However, the longer mice were on the linear section, the greater the deviation when reaching the bend. This suggests that mice form a moment-by-moment

expectation of the trail heading, integrating their past experience. The longer they are on a linear trail, the more confident they are about the future trail heading, which presumably helps them track linear sections better, but causes greater error when the trail deviates significantly from linearity. This finding offers further evidence of the use of predictive strategies by mice to navigate. Finally, we asked whether higher order features of the trail could influence tracking strategies in mice. We watched mice follow straight trails after a period of following convoluted trails of different complexities. Remarkably, mice followed identical linear stretches of odor trails with varying degrees of lateral deviations, depending on their recent experience – when following highly tortuous trails, mice seem to expect trails to deviate from linearity and predictively make larger lateral movements, leading to less accurate tracking of linear trails. Such experience-dependent differences cannot be accounted for by simple reactive trail tracking strategies.

We used a Bayesian filtering framework to build an agent-based model for predictive trail tracking. In the absence of detailed knowledge about the neural circuits involved in this behavior, we chose to frame the agent's memory of the trail in a general way using a Gaussian Process Regression (GPR). In such a formulation, the agent's estimate of the trail is obtained from a finite number of previous trail encounters. At any moment, the agent has an estimate of the future trail heading and the uncertainty associated with it. This is closely related to the sector search strategy (Reddy et al., 2022b), which uses Bayesian filtering to maintain a posterior probability distribution over possible trail headings by integrating past detections and non-detection, and where the agent then searches within a sector with high posterior probability. Here, we only use a finite memory of trail encounters, to enable flexible interpolation and explicit uncertainty quantification. Although a model with reactive turns and simple handcrafted extrapolations allows tight tracking for simple trail geometries, it fails for more complex trails that mice can track well. An agent that remembers up to 10 previous detections (memory time constant of 10-20 sec) can successfully track tortuous trails as well as those with breaks. Interestingly, such memory also leads to less tight tracking of straight trails immediately following adaptation to tortuous trails. While this positive comparison is intriguing, it should not be taken as proof that the animal is using (approximate) Bayesian filtering, which would require a more comprehensive examination and further data. The ability of an agent-based model with memory to describe the mouse performance much better than previous simpler models offers strong evidence for the existence of predictive strategies.

Why are these short-term adaptations to the trail geometry helpful to mice? In sensory systems, adaptation to the statistics of stimuli increases information transmission in a predictive manner (Attneave, 1954; Barlow, 1961; Laughlin, 1981; Manookin and Rieke,

2023). In our experiments, we observed adaptation in the complete sensorimotor behavior. After a significant period of experience with straight trails, mice might benefit from reinforcing the prediction of future linear heading to lessen computational load and increase tracking speed. Similarly, adapting to sinuous trails with more elaborate lateral movements might improve trail tracking efficiency without having to update trail heading as often. These adaptations, while helpful when the trail geometry remains unchanged, lead to more errors when the geometry abruptly changes, as in our experimental design. Such adaptation can be rationalized as building of internal models of the trail geometry to allow predictive strategies for more accurate tracking (Fiser et al., 2010; Kawato, 1999; Rao et al., 2023; Todorov and Jordan, 2002).

The neural underpinnings of the different features of trail tracking we have uncovered remain to be discovered. The sniff-triggered sensory representation of odor information from the two nostrils could be integrated in regions such as the anterior olfactory nucleus and the piriform cortex (Kikuta et al., 2010, 2008; Rabell et al., 2017; Wilson, 1997). These signals could then drive nostril and head reorientation towards the trail through collicular pathways (Rabell et al., 2017), followed by locomotor adjustments. An important aspect of the behavior is the modulation of such turning responses by previous history of detections. How sequences of detections and blanks are combined in the brain to obtain estimates of trail heading (goal or heading direction), so that appropriate corrections to head orientation can be made, remains unclear. The brain regions that estimate the direction of travel must also adapt to the statistics of the trail. Neural perturbations and recordings in candidate brain regions are needed to uncover these computations.

Materials and Methods

Animals: 42 adult Black6/J (000664) mice of both sexes were used in this study. All animals were obtained from Jackson Laboratories and maintained within Harvard University's Biological Research Infrastructure (BRI). Mice were housed in an inverted 12 h light cycle and fed ad libitum. Within the BRI, mice were group housed at 22 ± 1 °C at 30–70% humidity. Animals were 2–8 months old at the time of the experiments. Each behavioral session represents a unique animal throughout the experiments described here and multiple sessions have not been combined for any condition. All experiments were performed in accordance with the guidelines set by the National Institutes of Health and approved by the Institutional Animal Care and Use Committee at Harvard University.

Behavioral apparatus: Mice run freely on an 'endless' paper spool within an acrylic box (45cm x30cm) that extended close to the width of the paper roll (McMaster-Carr, 20565T13). Ink trails were delivered using an adapted inkjet printhead or delivered through a paint brush coupled to a stepper motor that was programmed to create movements using custom written code implemented on a PRJC Teensy 3.6 micro-controller (32 bit 180 MHz ARM Cortex-M4 processor). Video recordings were obtained using 940nm IR LEDs (LED Lightsworld), which is essentially invisible to mice. Images were recorded at 30 Hz by three FLIR Grasshopper 3, 4.1MP Mono USB3 Vision, CMOSIS CMV4000 cameras, one from above, one from the side and one from the left corner of the acrylic box. For all analysis in this paper, only videos from the top and side were used. The trail was imaged from outside the box using a FLIR BlackFly S, 3.1 MP, USB3, Sony IMX265 camera. The paper spool was driven by a powerful DC motor (ClearPath) and the displacement of the paper was obtained using a linear encoder coupled to a wheel placed on the paper. The control of the DC motor driving the treadmill was implemented on a separate control system run on a PJRC Teensy 3.2 micro-controller (32 bit 72MHz ARM Cortex-M4 processor). Ensure drops were delivered on the trail as rewards using a solenoid controlled by the Teensy controlling the ink delivery. The rewards were delivered across the entire width of the paper at random times, drawn from an exponential distribution. Data was acquired using Bonsai (v2.8.5, <https://bonsai-rx.org/> (Lopes et al., 2015)).

Respiration measurements: Respiration measurements were obtained through either wireless EMGs targeting the diaphragm — the main inspiratory muscle — or wired thermistors that were targeted to the olfactory epithelium. For wireless EMG recordings, the targeting was adapted from a procedure previously described (Hérent et al., 2020; Reisert et al., 2014). Briefly, mice were anesthetized using an intraperitoneal injection of ketamine/xylazine mixture (100 mg/kg and 10 mg/kg, respectively). Mice were placed in supine position, the peritoneum

was opened horizontally under the sternum, the sternum clamped to expose the diaphragm and a sterile parafilm was used to protect the upper part of the liver. The electrodes were then sutured in place on the diaphragm and the Stellar BTA-XS telemetry implants (Stellar Inc., Woodland Hills, CA) implant was placed in the subcutaneous pouch made close to the ventral abdominal area. The implant was sutured in place through the suture tab on the device to avoid movement. The implants were modified to have custom fine-wire electrodes to facilitate attachment to the diaphragm. Buprenorphine extended-release (3.25 mg/kg) (Ethiqa XR) and Carprofen (5 mg/kg) was administered subcutaneously for analgesia at the end of the surgery. In addition, Sulfatrim (80mg/kg) mixed with Ensure was provided to mice to prevent infections. The implants were calibrated to store the recordings in millivolts, and the sampling rate of 500 Hz was used. Data was recorded onto the device and streamed after every session. Thermistors were implanted following procedures described previously (Findley et al., 2021; McAfee et al., 2016). Briefly, mice were anesthetized, and lidocaine was applied before exposing the skull. The epithelium underlying the nasal bone was exposed by drilling a small hole and the bead of the thermistor was inserted halfway through. The thermistor was held in place was sealed using Kwik-cast and mill-max pins were secured on the skull using a custom, 3D printed half head bar. Ultralight flexible tethers were made by soldering two wires (Cooner wire, CZ 1187 wire, FEP Insulation 38AWG with 0.012" diameter) and the signal was amplified using a custom-built amplifier. The signal was sampled at 3KHz.

Nostril stitch: We closed one nostril of mice animals using 7–0 gauge MicroSutures (S-N718SP13). Mice were given lidocaine on the nostril and anesthetized using an intraperitoneal injection of ketamine/xylazine mixture (100 mg/kg and 10 mg/kg, respectively). The suture was pulled through the upper lip of the nostril, and the nostril was closed completely. The closure of the nostril was confirmed using a drop of water placed on the nostril, which was not aspirated. For sham stitches, the mice were anesthetized, and a suture was just looped and the nostril was not closed. Mice were administered extended-release buprenorphine (Ethiqa-XR) (3.25 mg/kg).

Commissure cuts: The anterior commissure was severed adapting an aseptic procedure described previously (Rabell et al., 2017). Briefly, mice were anesthetized using ketamine/xylazine and placed on a stereotaxic apparatus. Following a craniotomy, sapphire blade lancets (World Precision Instruments) were used (AP- -0.25 to 0.25, ML-0, DV-5) to sever the entire anterior commissure. Sham animals went through the whole surgical procedure except that the blade was only lowered to a DV of 4mm thus preventing transection. Extended-release buprenorphine (3.25 mg/kg) (Ethiqa XR) and Carprofen (5 mg/kg) was administered subcutaneously for analgesia at the end of the surgery

Histology: To visualize the transection of the anterior commissure, mice were deeply anesthetized with a ketamine/xylazine mixture and perfused transcardially with 10 mL of PBS (pH 7.4) first, followed by 10-20 mL of 4% paraformaldehyde in 0.1 M phosphate buffered saline (pH 7.4). After the brains were removed, 70-um thick horizontal sections were made using a vibratome (Leica). Slices were then washed and mounted for confocal imaging with DAPI mounting media and imaged with a confocal microscope (LSM 980, Zeiss).

Trail tracking agent and Gaussian Process Regression (GPR): A model of an agent tracking a trail was implemented. Fictive odor trails were generated as piecewise continuous functions, constructed from a sequence of linear and random (normally distributed) segments. Trail curves were interpolated via cubic splines over a fixed number of points (40 for low tortuosity and 80 for higher tortuosity), and breaks in specific conditions were introduced. Three types of agents were simulated viz. 1) a completely reactive agent, 2) An agent with limited memory and using odor gradient information to make estimates of the trail, similar to the model presented in (Khan et al., 2012), and 3) an agent using a moving memory window of the M most recent contacts with the trail fitting a GPR model. The implementation uses the GPR class from Scikit-learn v1.3 (Pedregosa et al., 2011)

For a reactive model, the agent ignores all past crossings and predictions, executing only sniff-by-sniff reversals upon loss of odor or high lateral error, mimicking a pure reactive implementation with no memory of the odor trail or concentration. For the limited memory agent, we implemented the model from (Khan et al., 2012) with the exact parameters from in their supplementary methods. Briefly the agent advances along the trail at a fixed forward velocity, performing lateral oscillations by alternating the sign of an imposed “casting” acceleration. At each time-step, after an “inhalation”, the agent’s position was updated using: $x_{t+1} = x_t + v_x \Delta t$ and $y_{t+1} = y_t + v_y \Delta t + \frac{1}{2} a_y \Delta t^2$. A reversal of the casting acceleration is either because the odor gradient is greater than the prescribed threshold or the concentration detected drops below the assumed threshold or because the lateral deviation from the current trail estimate surpasses the current uncertainty bound. The agent maintains a running window of the most recent (up to the past 5) trail crossings and maintains an internal estimate of the trail slope.

For the agent with the Gaussian Process Regression (GPR) based on encountered trail contacts, when at least two contacts were obtained, the agent fits a GPR model to recent crossing points (x_k, y_k, t_k) , using (x_k) as inputs. In pseudocode, the GPR prediction and

posterior uncertainty at position x was given by: $[\hat{z}(x), \hat{\sigma}(x)] = GP.predict(x)$. The error estimate of the trail was defined by the GPR prediction $\pm k$ posterior standard deviations with k being 2. A Matern kernel with a smoothness of 1.5 was initialized with a length scale of 5-500mm before the fit was optimized. The choice of the kernel and the exact parameters are not critical since the model is a phenomenological one designed to recapitulate behavioral data from mice. At every trail contact the GPR was re-fit with the data points weighted by recency given by $w_i = \exp(-\lambda(t_{last} - t_i))$. In addition, since the standard implementation of the scikit-learn GPR did not support weighting samples, an exponential weight of trail contacts was created by replicating each crossing in the training set proportional to w_i . Using such an approach more recent trail contacts are weighted more than temporally older trail contacts. Each agent simulation was performed in batches of 30 independent replicates per condition (which included trail statistics and the four recency weights). For testing the effects of different recency weights, an additional batch of 50 simulations were performed. Each simulation was initialized using a random seed. A memory of 20 was used for the GPR agent. All measurements were performed after memory of the agent was saturated with 20 trail contacts so that the internal estimate obtained by the GP fit had stabilized.

Data analysis and Statistics: Markerless pose estimation was performed using DeepLabCut v2.2.0 (Mathis et al., 2018; Nath et al., 2019) to obtain body parts of interest, viz. the snout (the tip of the nose), the ears, the shoulder and the base of the tail. For the view from above, 950 frames taken from 36 videos for 350,000 training iterations with ResNet-50v1 and the train and test errors were 6.69, 7.85 pixels respectively with a 0.1 p-cutoff (2048 x 1400 image size). For the view from the side, 380 frames taken from 8 videos for 150,000 training iterations with ResNet-101 and the train and test errors were 5.91 and 20.03 with a 0.1 p cutoff (2048 x 760 image size). The trail was also detected using DeepLabCut to identify the edges of the trail, and other features such as rewards. 3000 frames taken from 40 videos were used with ResNet-101 for 450,000 training iterations and the train and test errors were 21.63 and 24.53 with a p-cutoff of 0.1 (1504 x 608 image size). Sessions were reconstructed by “rolling out” using an odometer that used a rotary absolute shaft encoder (US digital MA3) connected through a wheel placed on the paper. The encoder was read out using the same microcontroller controlling the DC motor and was interfaced with the acquisition computer using serial communication. For estimating the elevation of the snout, another network was trained to detect the limbs of the animal. For this, 670 frames taken from 17 videos for 200,000 training iterations with ResNet-50v1 and the train and test errors were 2.6, 12.36 pixels with a 0.1 p-cutoff (2048 x 760 image size). The training was done using a NVIDIA Titan V or a 2080Ti GPU. After obtaining the x, y coordinates of their hands and feet, a straight line was approximated to obtain the fore-hind paw plane using RANSAC (RANDOM SAMPLE

Consensus) robust linear regression implementation to reduce the sensitivity to the outlier points. The orientation (slope and intercept) of this fitted line provided an estimate of the mouse's base of support at that instant. Finally, the vertical intersection between the x coordinate of the snout and the fitted line was computed and plotted back on the frame for manual verification.

For estimation of respiration, signals were first filtered using a Butterworth filter to remove high frequency noise, following which a Hilbert transform was applied to obtain the envelope. Peaks and troughs were then detected using the *find_peaks* function in SciPy (Virtanen et al., 2020).

No statistical method was used to predetermine sample size. Statistical details including tests and p-values can be found in the figure legends and results section. Number of animals are reported in the text or the figure legend. In the text, we report mean $\pm$ s.e.m. (standard error of the mean) for datasets where means were compared. Jackknife estimation was used to estimate errors in percent tracking, by leaving out each individual animal.

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Author Contributions

S.J., A.M., M.V. and VNM conceived the project. A.M. designed and built the initial experimental setup and piloted experiments with M.W.M. S.J. refined and adapted the behavioral rig, collected and analyzed the data reported in this paper, with input from N.R., A.M., M.V., and V.N.M. S.J. and N.R. implemented the model with inputs from M.V. and V.N.M. S.J. and V.N.M. wrote the manuscript with edits and inputs from all the authors.

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Supplementary Figures

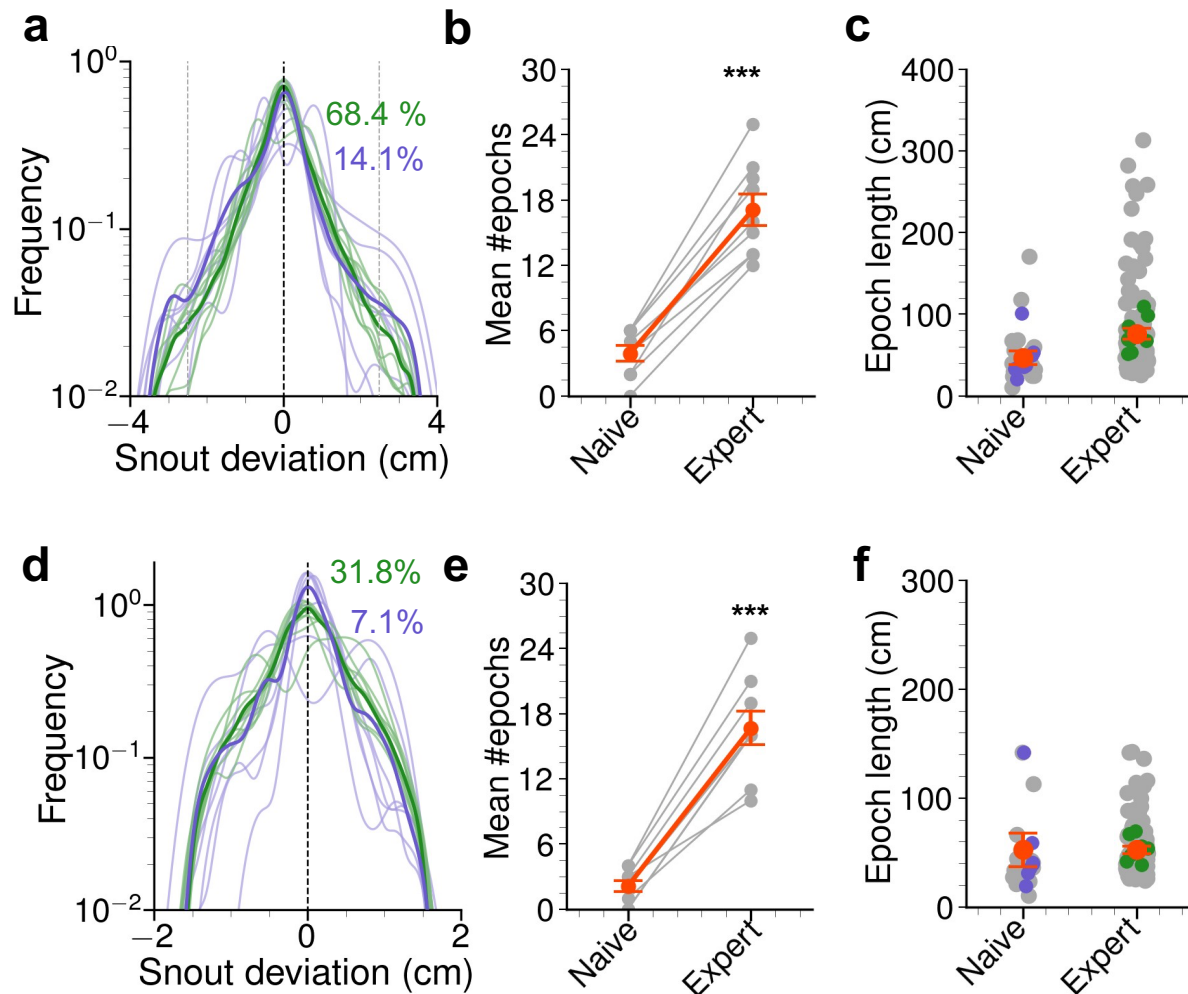


Figure S1: Trail tracking differences across learning are not influenced by selection criteria. Parameters of following a trail with two selection criteria that are different from that shown in Figure 1d. 3.5 cm for 2.5 seconds in panels a-c, and 1.5cm for 2.5 seconds in panels d-f. Each line represents a separate animal (a) The same distribution shown in Figure 1c limited to 3.5 cm on either side of the trail. Distributions of the deviations of the snout from the trail. Numbers indicate percentage tracking. (b) Mean number of epochs. Gray points represent 9 individual animals and red is average. (c) Mean length of epochs. The colored dots represent individual animals. (d) The same distribution shown in Figure 1c limited to 1.5cm on either side of the trail. Numbers indicate percentage tracking. (e) Mean number of epochs. (f) Mean length of epochs. These are the same animals shown in panel d-f of Figure 1 (9 animals).

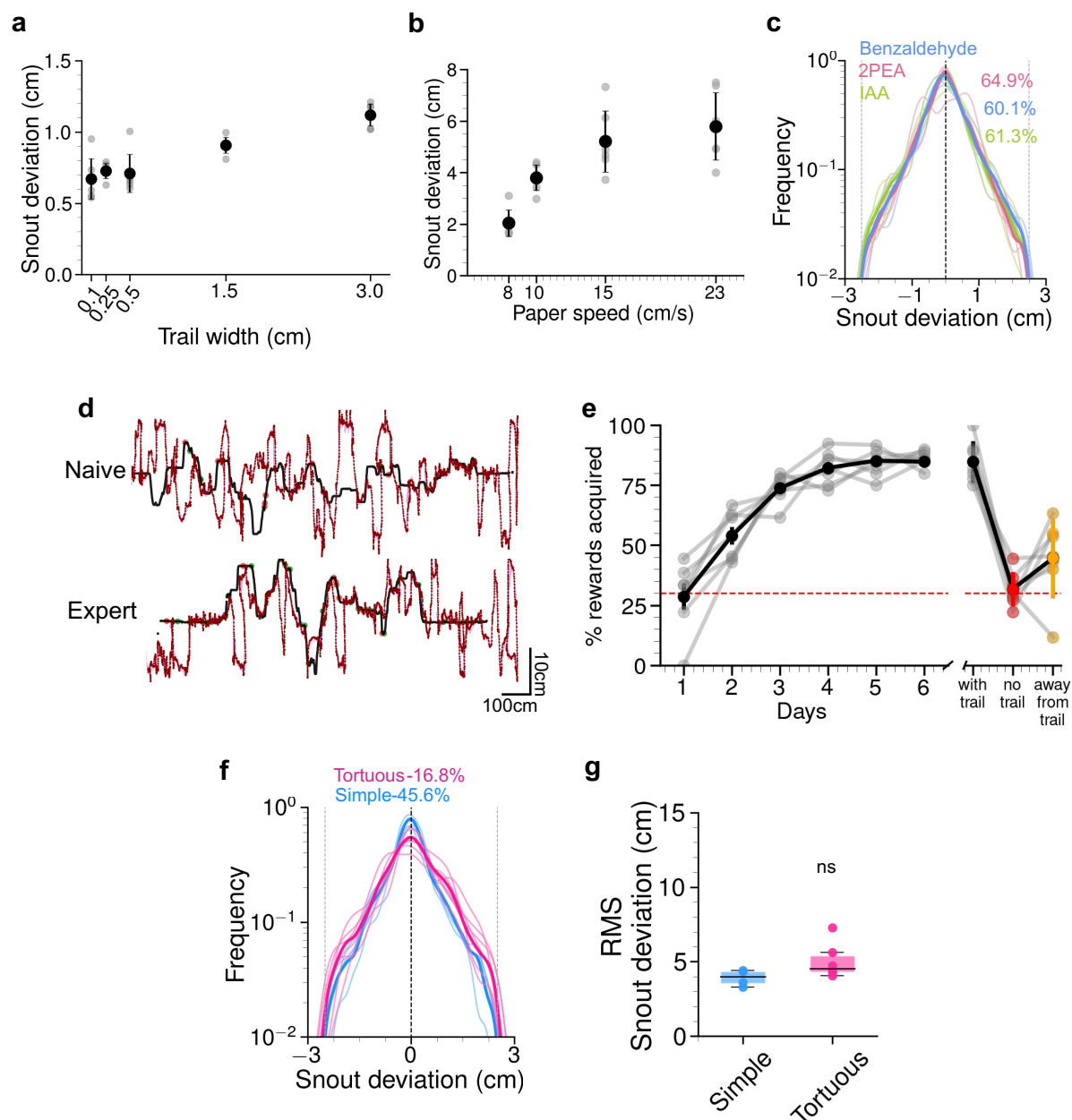


Figure S2: Factors that influence trail tracking. (a) Average deviation from the trail from different sessions with varying widths of the trail displayed as mean $\pm$ s.e.m (6 animals). Each gray dot is a separate animal. (b) Average deviation from the trail as a function of the speed of the paper displayed as mean $\pm$ s.e.m. Each gray dot is a separate animal. (c) Distributions of the deviations of the snout from the trail from mice that were presented with different odorants. The numbers indicate percentage tracking. (d) Example session from a mouse following a trail on day 1 (naïve) and on day 6 (expert). (e) Percentage of rewards consumed while following odor trails over days shown as average $\pm$ s.e.m. (7 animals) (f) Distributions of the snout deviations from the trail of mice that were following simple or tortuous trails. (g) Root mean square deviations from the trail from animals tracking either a simple straight trail or a tortuous trail (ns: not significant, $p = 0.17$, Mann-Whitney U test, 5 animals).

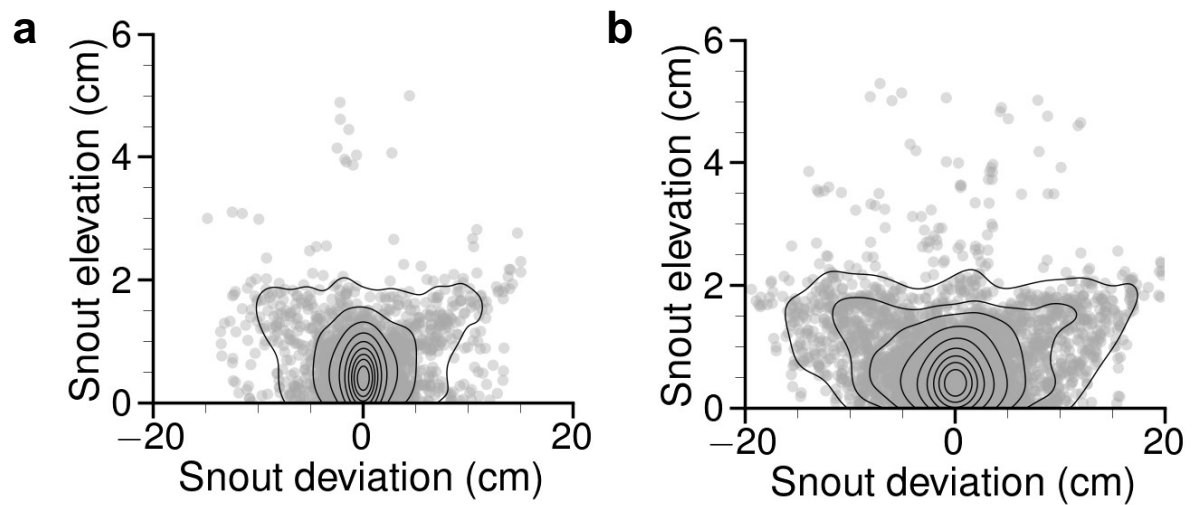


Figure S3: Introduction of the trail odor in the air disrupts trail tracking. Distributions showing snout occupancy in 3d space without (a) and with (b) the trail odor presented aurally. The distribution is combined from 9 animals.

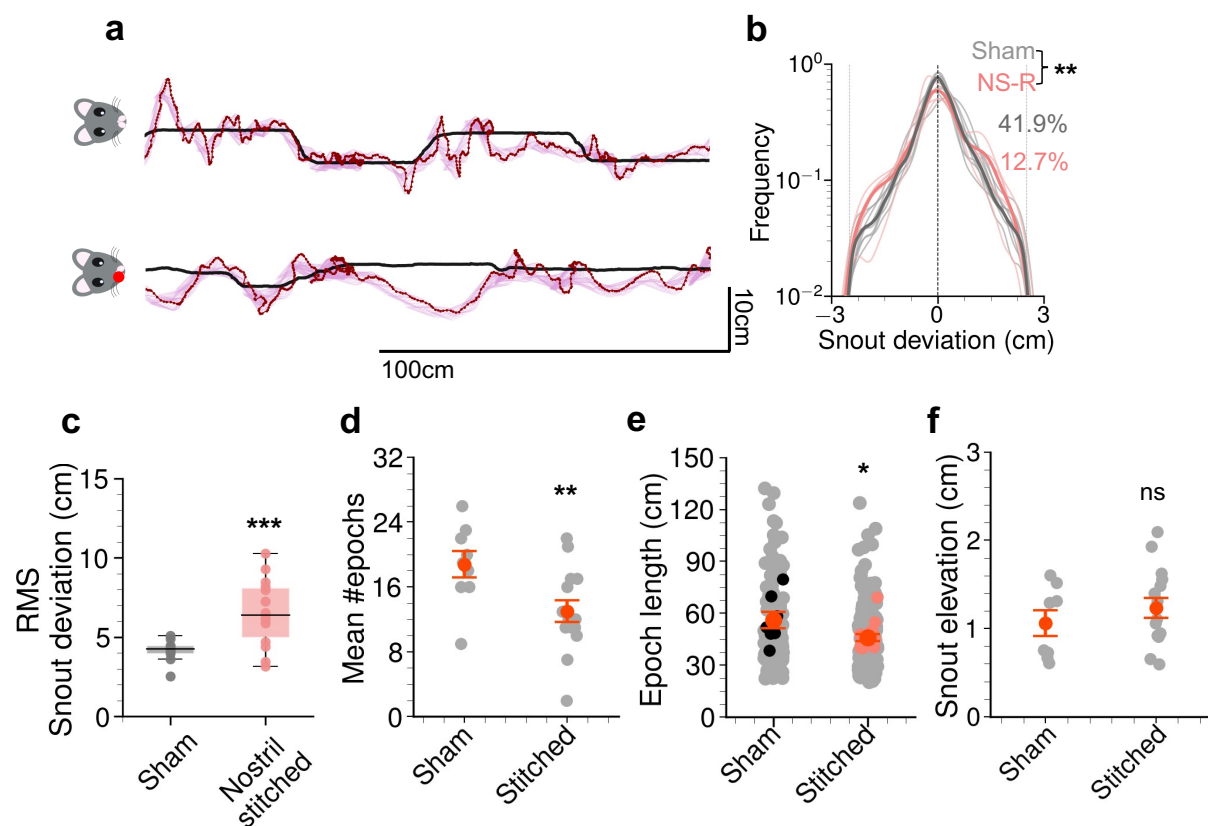


Figure S4: Blocking the nostril disrupts trail tracking. (a) Example trajectory from a mouse following a trail after sham stitch (see Methods) or when the right nostril is stitched. (b)

Distributions of the deviations of the snout from the trail from mice with a sham stitch and when the right nostril was stitched. Numbers indicate percentage tracking. Individual lines show each animal (**c-f**) Performance metrics of trail tracking for sham and mice with a single nostril stitched. Individual dots in c,d and f represent unique animals as do the colored dots in e. (**c**) Root mean square deviation from the trail from either sham or nostril stitched conditions. (**d**) Mean number of epochs. (**e**) Mean length of epochs and (**f**) Elevation of the snout from the paper floor. (** $p < 0.01$, ns: not significant, Wilcoxon rank-sum test).

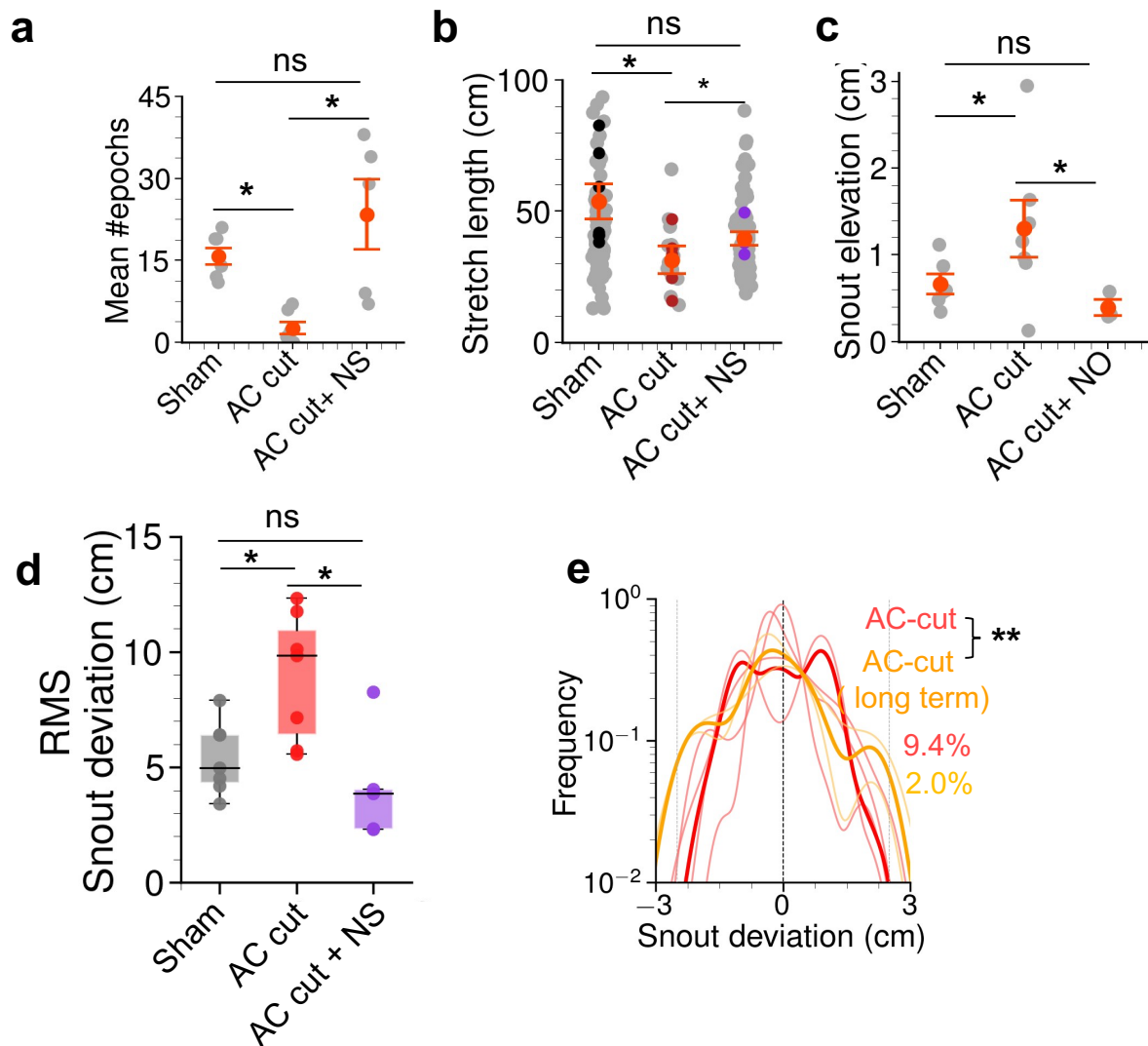


Figure S5: Summary of trail tracking measures after the anterior commissure was severed. (a-c) Performance metrics of trail following between sham, mice with anterior commissure severed and with anterior commissure severed but with one of the nostrils stitched. Gray dots in a,c as well as colored dots in b represent unique animals. (a) Mean number of epochs. (b) Mean length of epochs and (c) Elevation of the snout from the paper floor. (d) Root mean square deviation from the trail from either sham, animals with the commissure transected or animals with the anterior commissure transected and one of the nostrils stitched. (e) Distributions of the deviations of the snout from the trail from mice where

the anterior commissure was severed immediately after (red) and after 2 weeks (orange). Lines represent individual mice (** $p < 0.01$, * $p < 0.05$, ns: not significant Wilcoxon rank-sum test). Numbers indicate percentage of time mice spend tracking (as defined in Methods).

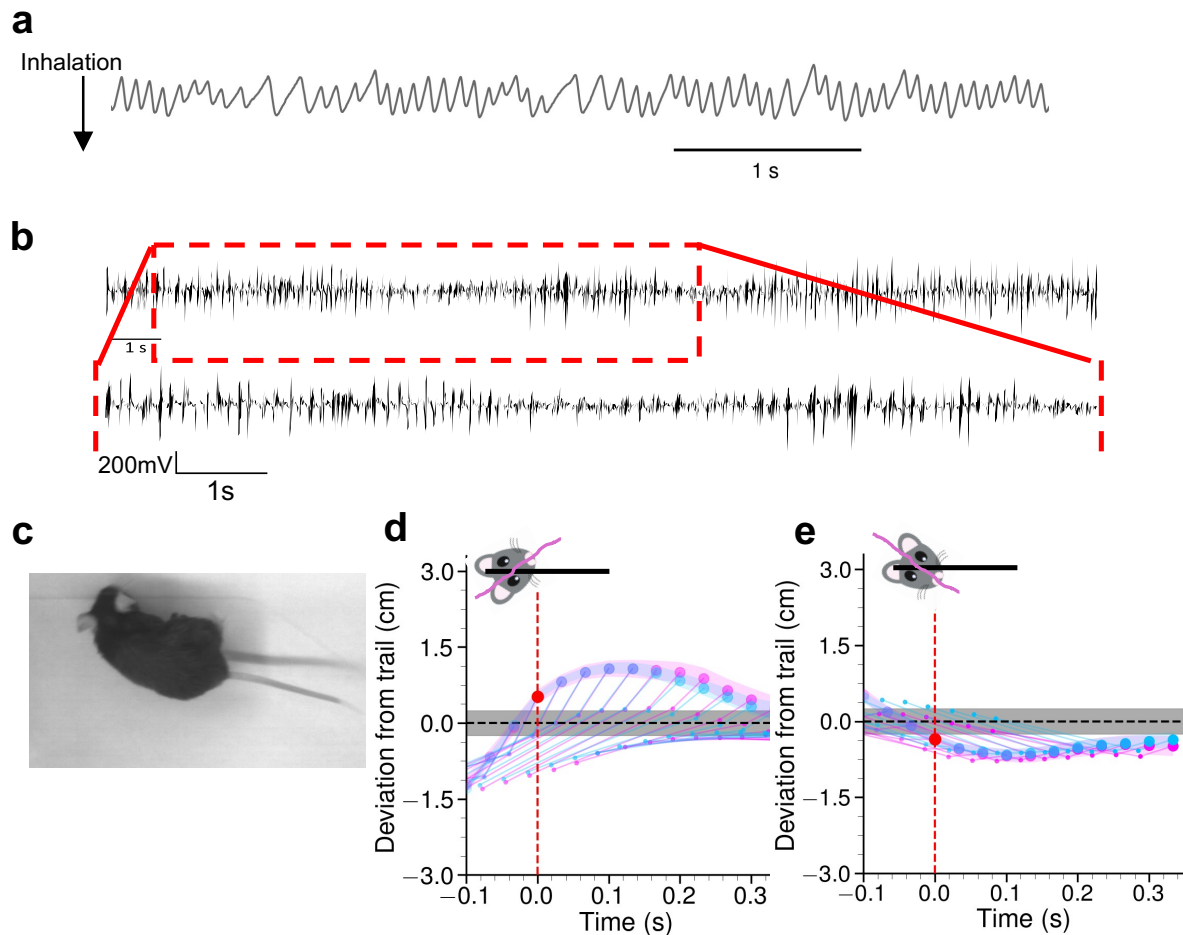


Figure S6: Measuring respiration during trail tracking. (a) Example time series from a thermocouple from a mouse following the trail. (b) Example EMG trace from a mouse implanted with a wireless EMG implant, with the inset showing a portion of the trace at higher resolution. (c) Image showing a mouse implanted with the wireless EMG sensor following a trail. (d and e) Trajectories showing snout, shoulder and base of the tail, joined by lines. The red dot indicates an inhalation event. Pink traces are from when an inhalation was detected at $t = 0$, and cyan traces are when there is no inhalation at $t = 0$. The trail is 0.5cm in width and shown as a gray line. The distance at the inhalation is between 0 and 1cm and is on the trail.

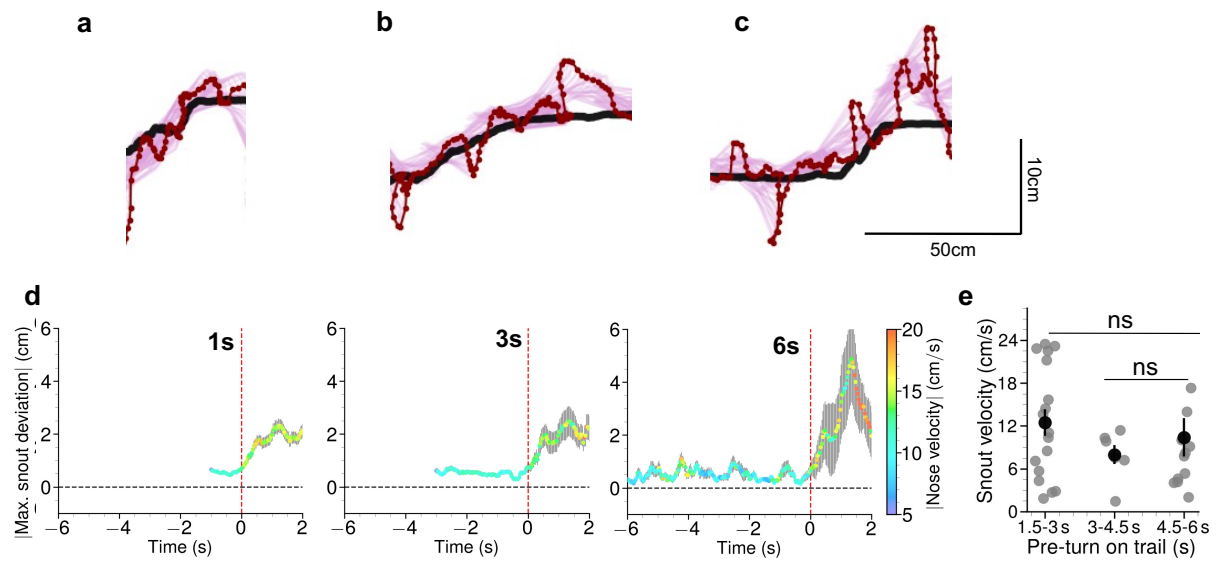


Figure S7: Off times for memory post encountering noisy trails. (a,b,c) Example trajectories from a mouse encountering a turn in the trail, after following the trail for 1s (a), 3s (b) or 6s (c). (d) Snout positions over time shown in Figure 4e colored by nose velocity. (e) Summary showing the average velocity of the nose at the point when the trail turns, binned into 1.5 second bins. These are not significantly different ($p > 0.8$, Kruskal-Wallis ANOVA with post hoc Mann-Whitney U tests using Bonferroni multiple comparisons correction).

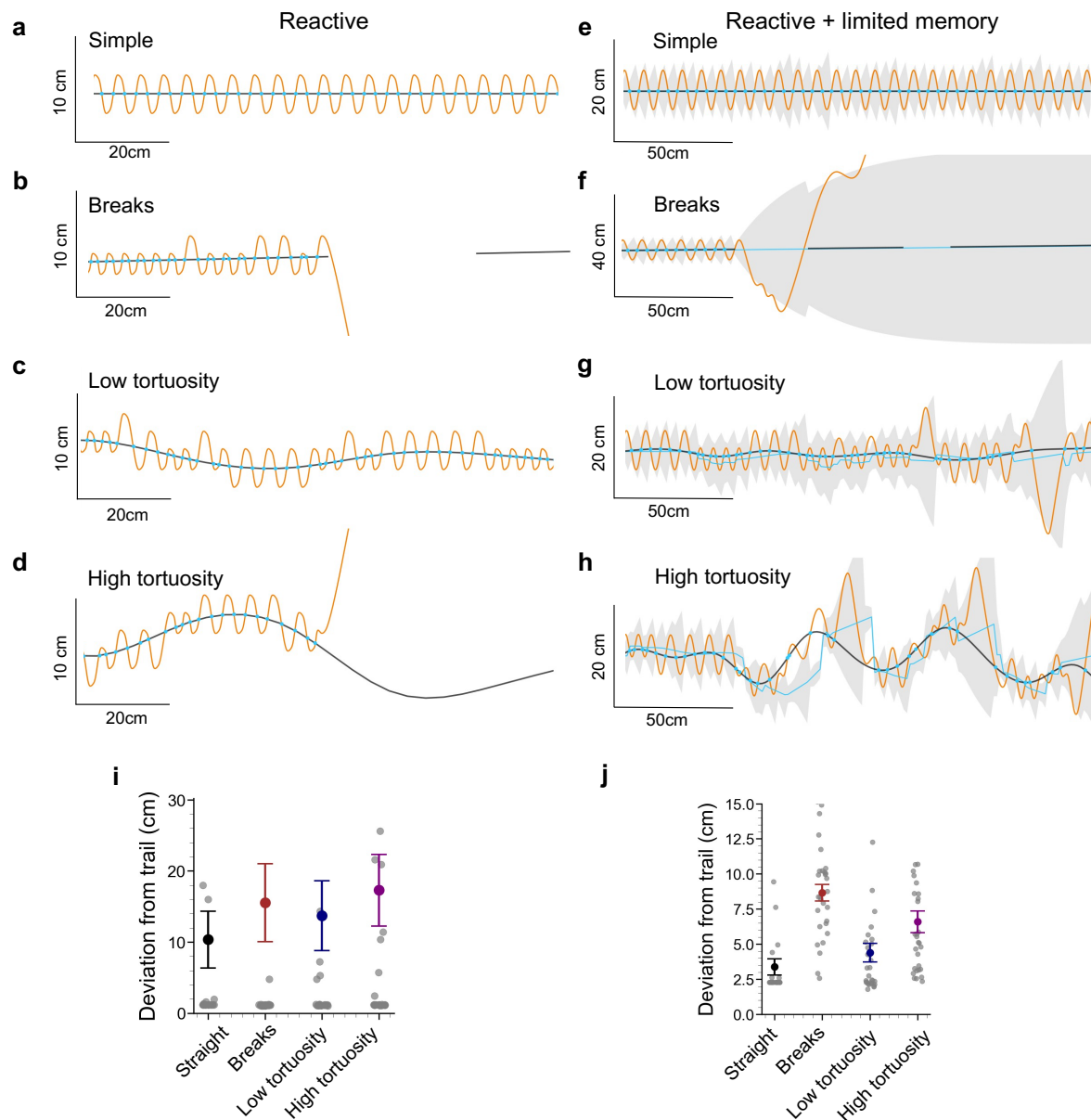


Figure S8: Reactive models and models with limited memory can track trails with limitations. (a-d) Example trajectories of a reactive agent with no memory following a trail that is straight (a), simple with breaks (b), of low tortuosity (c) or high tortuosity (d). (e-h) Example trajectories of an agent that is reactive but also has up to 5 previous contact points in memory to form an estimate of the trail that was presented a trail that is straight (e), simple with breaks (f), of low tortuosity (g) or high tortuosity (h). (i) Snout deviations from 30 simulations across the various trail types shown through a-d. (j) Snout deviations from 30 simulations across the various trail types shown through e-h.

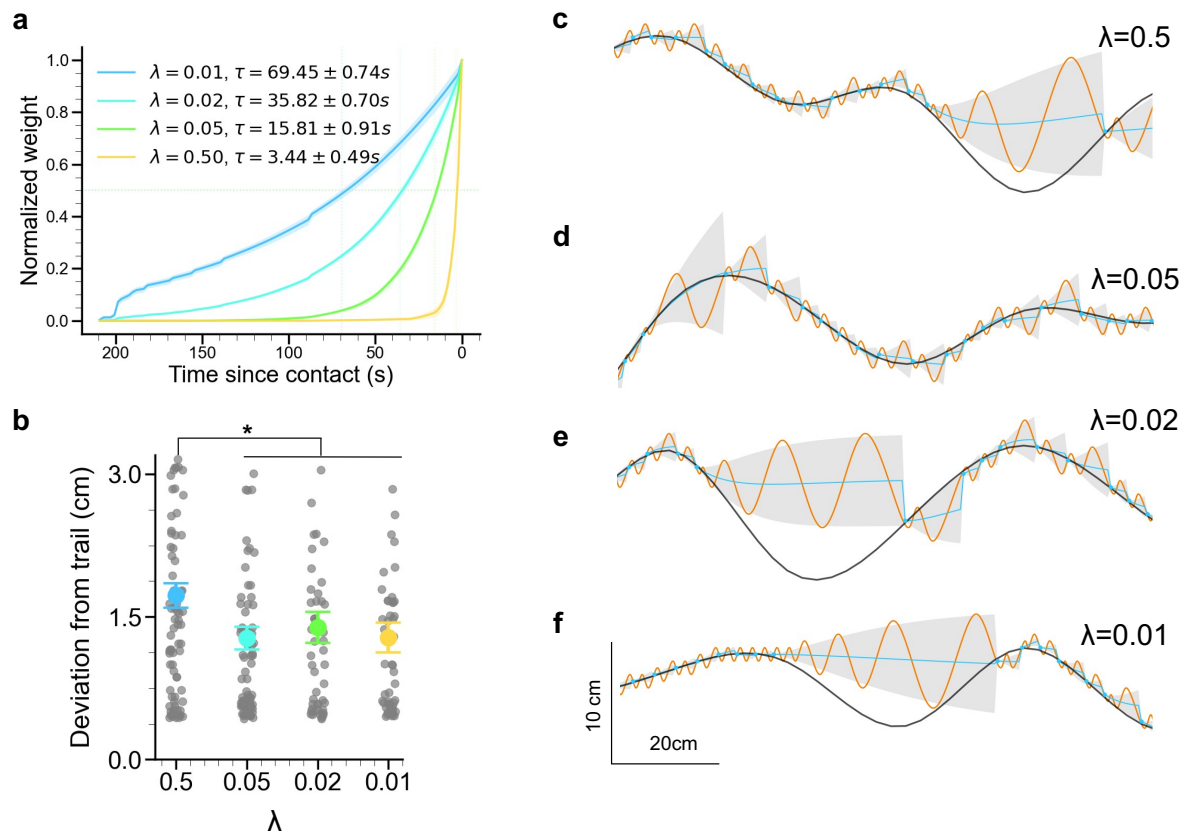


Figure S9: Agents with memory. (a) Normalized weight of previous contacts plotted against time since last contact from 30 agent simulations for the same four values of λ shown in (c-f). The most recent contact is on the right. (b) Deviations from the trail shown as mean $\pm$ s.e.m from at least 80 simulations of agents following a trail of high tortuosity for the same four values of λ . The snout deviation of the agents with a λ value of 0.5 is significantly higher than $\lambda = 0.05, 0.02$ or 0.01. * $p < 0.05$, Kruskal-Wallis ANOVA with post hoc Mann-Whitney U tests using Bonferroni multiple comparisons correction test. (c-f) Example trajectories from an agent following a fictive trail of high tortuosity with varying lambdas λ that determine how many past contacts are weighted.